Individual Project Instructions and Examples

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October 19, 2026

CTS 2000: Scripting for the Humanities

Individual Project - Motivations

To build a strong foundation in digital humanities, you need to become familiar with how digital humanists use Python in their research. This involves studying and analyzing code written by other researchers, as well as reproducing their workflows to understand both the technical and interpretive dimensions of their work. It also means critically evaluating the projects you study—examining their data sources, methodological approaches, design choices, and coding decisions—to assess their scholarly rigor, transparency, and impact.

Individual Project – Goals

- **Familiarize** yourself with how digital humanists use Python
- Study and analyze **other researchers' code** and reproduce it
- **Critically evaluate digital humanities work**, its data, sources and approaches as well as its design and coding decisions
- Practice **presenting** with Jupyter notebooks and **giving feedback** to your peers
- May support the **literature review** for your group project (not required)

Individual Project - Components

Th individual project consists of two parts:

1. Short Paper
2. Jupyter Notebook Presentation

Short Paper

Due October 30, 2026

You will choose a topic, either a methodology such as topic modeling or a task such as named entity recognition, and identify **three papers on that topic**. You will briefly summarize their methods, and the reasons the authors chose them, and study their code and data. You will then critically evaluate the papers to identify their weaknesses and strengths. Your short paper should be 2 to 3 pages long (ca 500-750 words).



Code Presentation

Due November 4, 2026

You will replicate the code from one of the papers you evaluated and walk us through it. To do this, you will create a Jupyter Notebook and get the research code running in our course environment or on Google Colab. You will record yourself running the code and explaining each step to an audience unfamiliar with your topic but familiar with everything that we have covered in this class. Your presentations should be 5-6 mins long. You will upload your presentation and your Jupyter Notebook to CourseLink.



Code Presentation

Due November 4, 2026

Once you upload your presentation, **you will be matched with two other students**. You are expected to watch their presentations carefully and ask constructive questions. We will discuss how to give feedback and clarify the details about the matching process later in the semester. The quality of your engagement with your colleagues' work is an essential component of this assignment.



Reproducing code deepens understanding, sharpens technical skills, and reveals research methodologies.

Where and how to find code and projects?

Resources



General

https://aclanthology.org/
https://www.nature.com/palcomms/
https://paperswithcode.com/
https://zenodo.org/
https://scholar.google.com/
https://www.semanticscholar.org/

Digital Humanities

https://www.digitalhumanities.org/dhq/
https://euppublishing.com/loi/ijhac
https://openhumanitiesdata.metajnl.com/
https://academic.oup.com/ds
https://culturalanalytics.org/
https://journalofdigitalhistory.org/

Please choose a topic that is relevant for digital/computational humanities. You can browse through *Reviews in DH* for some inspiration.

Reviews in DH

https://reviewsindh.pubpub.org/

Reviews in Digital Humanities, edited by Dr. Jennifer Guiliano and Dr. Roopika Risam, is the pilot of a peer-reviewed journal and project registry that facilitates scholarly evaluation and dissemination of digital humanities work and its outputs. We accept submissions of projects that blend humanistic and technical inquiry in a broad range of methods, disciplines, scopes, and scales. These include but are not limited to: digital archives, multimedia or multimodal scholarship, digital exhibits, visualizations, digital games, and digital tools.





Example

Short Paper:
This is a mini *literature review*

General problem/task definition: What are these papers trying to solve, and why?
Concise summaries of the articles: Do not simply copy the article text in full. We can read them ourselves. Put in your own words the major contributions of each article.
Compare and contrast: Point out the similarities and differences of the papers. Do they agree with each other? Are results seemingly in conflict? If the papers address different subtasks, how are they related? (If they are not related, then you may have made poor choices for a lit review…).
Future work: Make several suggestions for how the work can be extended. Are there open questions to answer? How do the papers relate to your final project idea?
References section: The entries should appear alphabetically and give at least full author name(s), year of publication, title, and outlet if applicable (e.g., journal name or proceedings name). References are not a part of the page count.



DE GRUYTER Digit. Stud. in Lang. and Lit. 2024; 1(1–2): 113–141

Review Article

Defeng Li*, Kan Wu and Victoria L.C. Lei

Applying Topic Modeling to Literary Analysis: A Review

https://doi.org/10.1515/dsll-2024-0010
Received August 18, 2024; accepted November 6, 2024; published online November 26, 2024

Abstract: This article examines the integration of topic modeling within literary studies, highlighting its potential to transform conventional literary analysis through computational techniques. It reviews the theoretical underpinnings underlying topic modeling, including prominent algorithms like Latent Dirichlet Allocation, Non-negative Matrix Factorization, and Neural Topic Models, and discuss their utility in dissecting large textual corpora to uncover latent thematic and stylistic patterns. The article subsequently addresses the specific methodological steps for effective implementation, spanning text preprocessing model tuning and result interpretation. We further illustrate the diverse applications of topic modeling in literary studies through thematic analysis, comparative studies, and the extraction of cultural and historical insights. Challenges such as model accuracy, technical limitations, and ethical considerations are critically assessed. The review concludes by envisioning prospective future directions that foresee enhanced integration of topic modeling into literary criticism, facilitated by technological and interdisciplinary advancements.

Keywords: topic modeling; literary studies; computational literary analysis; latent dirichlet allocation; textual data analysis

https://doi.org/10.1515/dsll-2024-0010

Code Reproduction:

This is both a Jupyter Notebook submission and a presentation

Your goal here is not to do something new but to reproduce a piece of code that is interesting for you. You will learn a lot about how this code works and how you can adapt it to your work.



Computer Science > Computation and Language

[Submitted on 11 Mar 2022]

BERTopic: Neural topic modeling with a class-based TF-IDF procedure

Maarten Grootendorst

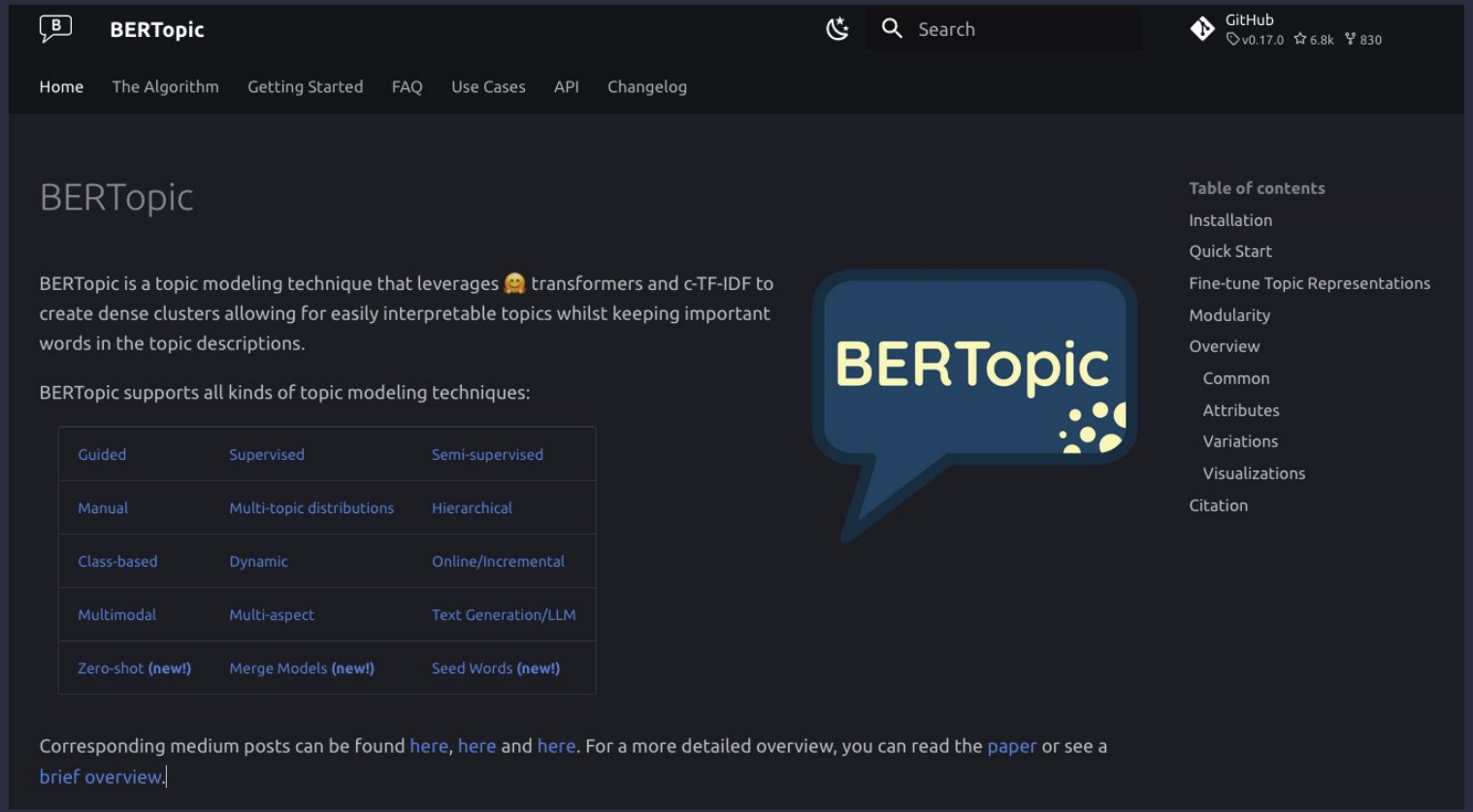
Topic models can be useful tools to discover latent topics in collections of documents. Recent studies have shown the feasibility of approach topic modeling as a clustering task. We present BERTopic, a topic model that extends this process by extracting coherent topic representation through the development of a class-based variation of TF-IDF. More specifically, BERTopic generates document embedding with pre-trained transformer-based language models, clusters these embeddings, and finally, generates topic representations with the class-based TF-IDF procedure. BERTopic generates coherent topics and remains competitive across a variety of benchmarks involving classical models and those that follow the more recent clustering approach of topic modeling.

Comments: BERTopic has a python implementation, see this https URL
Subjects: **Computation and Language (cs.CL)**
Cite as: arXiv:2203.05794 **[cs.CL]**
(or arXiv:2203.05794v1 **[cs.CL]** for this version)
https://doi.org/10.48550/arXiv.2203.05794

Submission history

From: Maarten Grootendorst [view email]
[v1] Fri, 11 Mar 2022 08:35:15 UTC (384 KB)

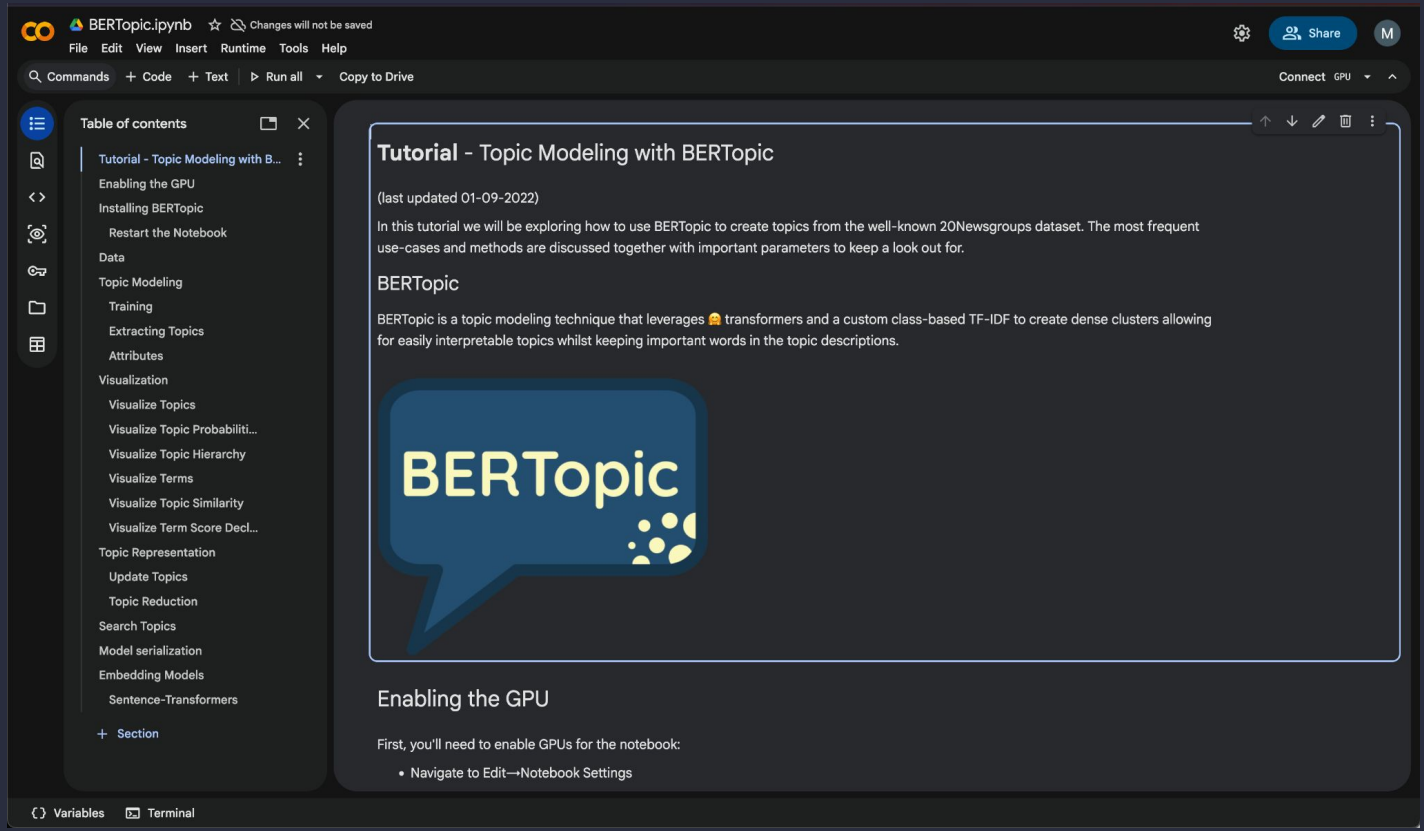
https://arxiv.org/abs/2203.05794



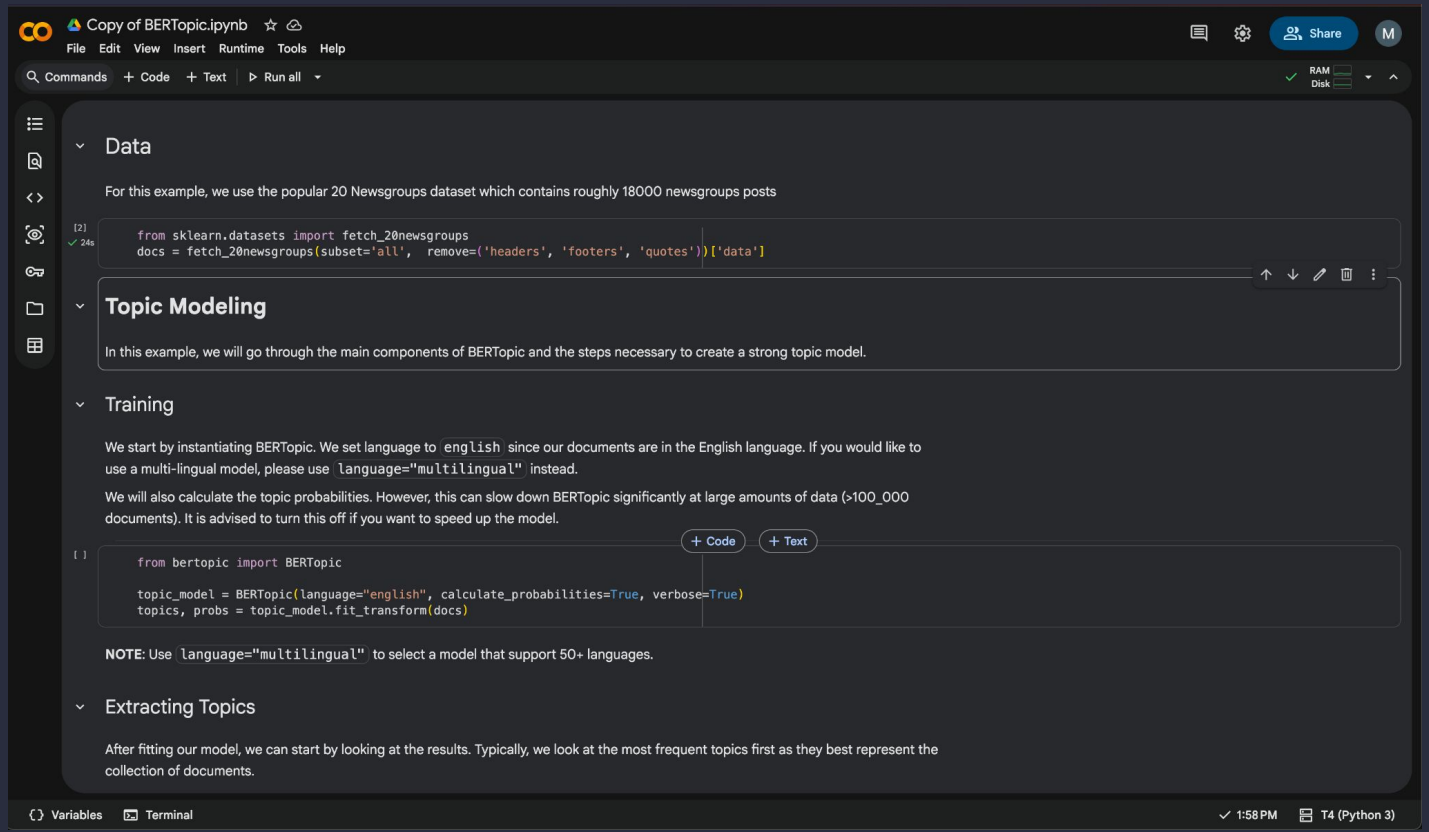
https://maartengr.github.io/BERTopic/index.html

Steps

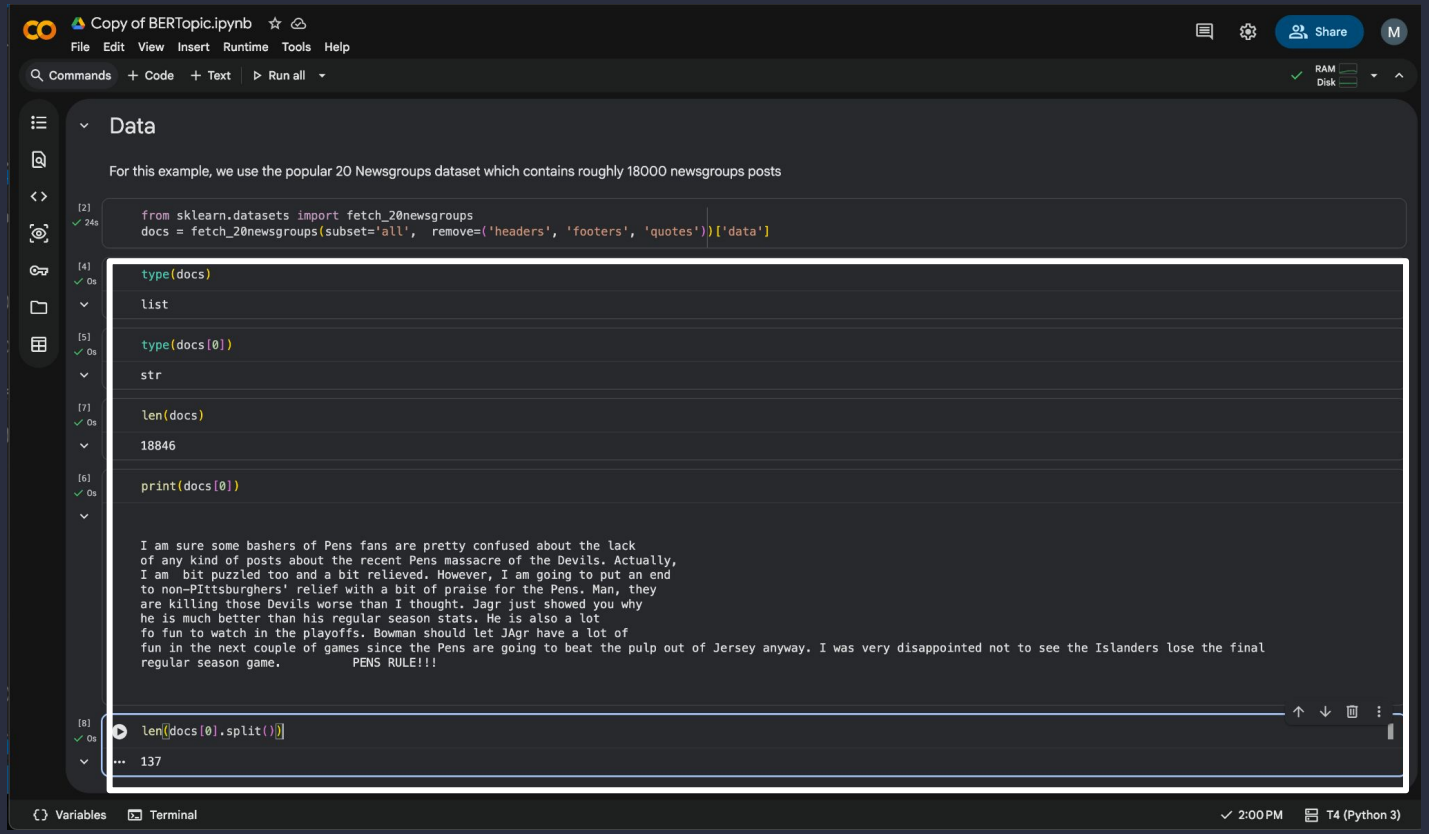
1. Open your VS Code or Google Colab and create a new Jupyter Notebook
2. Start installing necessary libraries and dependencies
3. Bring in any datasets that you need to run the code
4. Copy paste the code from the original source
5. Run it cell-by-cell, where necessary add explanations and even more code of your own. You want to see what is in the dataset, even if the author of the code does not have a line that explicitly shows it.



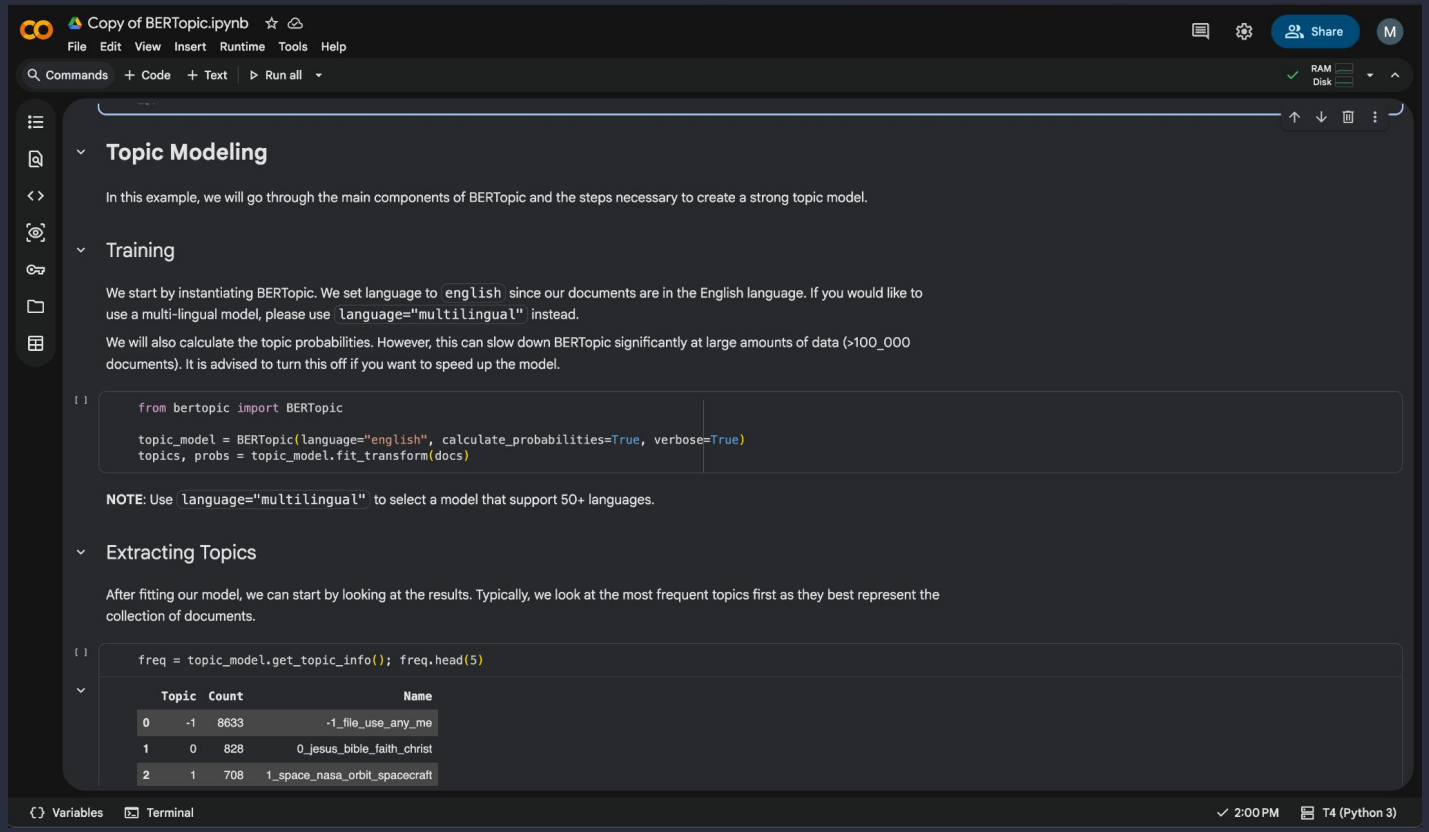
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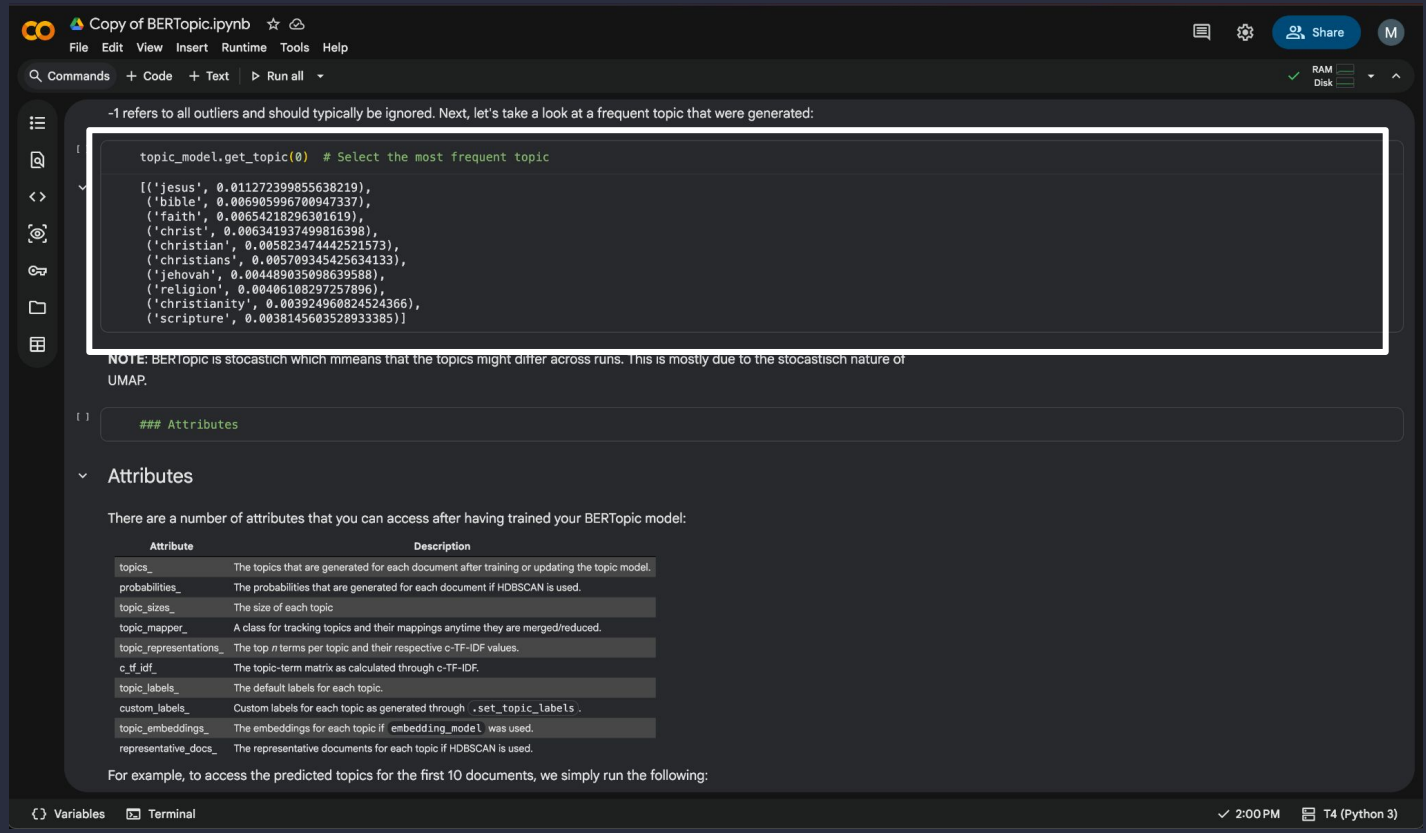
Let's load the data



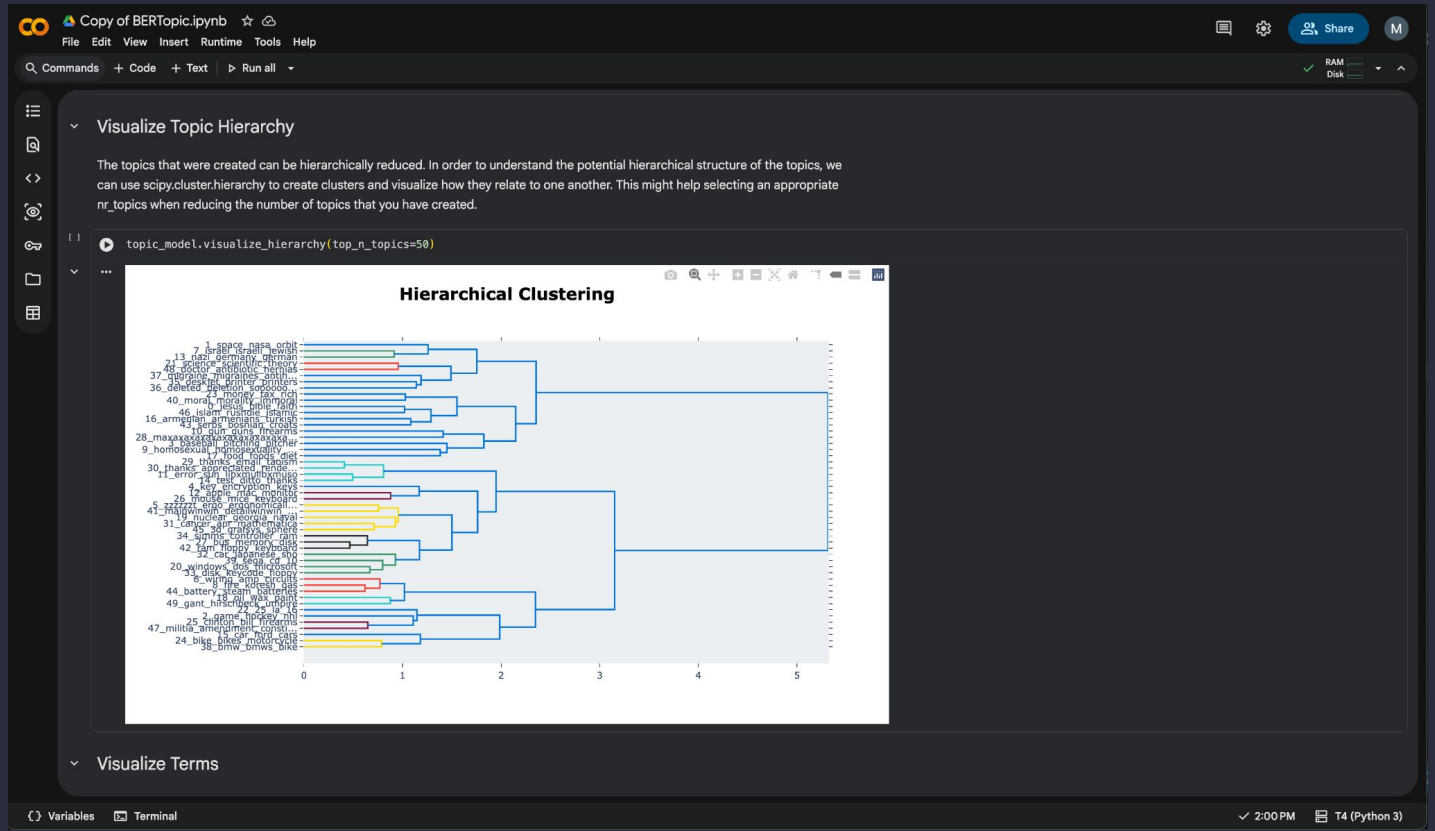
Expand upon the parts of the code that contribute to your presentation



Here, you need to tell us about what topic modeling is and what it does.



Explain what the outputs show


Copy of BERTopic.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
Visualize Topic Hierarchy
The topics that were created can be hierarchically reduced. In order to understand the potential hierarchical structure of the topics, we can use scipy.cluster.hierarchy to create clusters and visualize how they relate to one another. This might help selecting an appropriate nr_topics when reducing the number of topics that you have created.
topic_model.visualize_hierarchy(top_n_topics=50)
Hierarchical Clustering
1_space_nasa_orbit
7_israel_israeli_jewish
13_nazi_germany_german
21_science_scientific_theory
48_doctor_antibiotic_hermas
37_migraine_migraines_anoth...
14_printer_printers_
36_deleted_deletion_sroyya
40_moral_morality_objective
0_jesus_christ_bible
46_islam_rushdie_islamic
16_armenian_armenians_turkish
43_serbs_bosnian_croats
10_gun_guns_firearms
28_maxaxaxa

Any questions?